

Web Technology I

Unit 4 - HTML Editors and Tools

Unit 4. HTML Editors and Tools [5 Hrs.]

4.1. Introduction to HTML Editors and HTML Converters

4.2. HTML Editors and tools

4.2.1. Use of different HTML editors and tools like Dreamweaver, Microsoft Front Page, Notepad++, etc.

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4.3.1. Use of different graphical and animation tools like Adobe Photoshop, MS Paint, Flash, etc.

4.3.2. Adding Sounds and Animation to the web page (using embed tag)

We all know that programming languages require editors to write and execute code. For HTML, these code editors are HTML editors that help developers create and edit HTML documents.

4.1. Introduction to HTML Editors and HTML Converters

HTML Editors -

An HTML editor is computer software used to write and edit HTML code. This type of software helps manage the entire web development project by using different modes of document editing and functionality. Using text-based HTML editors, users can edit the source code directly, and WYSIWYG (what you see is what you get) editors can display documents for editing in the same way a web browser will show them afterward. These editors are usually part of a particular integrated development environment.

Example - Sublime Text, Atom, Notepad ++, Visual Studio Code, Adobe Dreamweaver, Brackets NetBeans.

Note –

Common HTML text editors are Notepad (PC) or TextEdit (Mac). Users can use these editors to write and execute HTML code to learn and practice HTML; we recommend users to use these simple text editors for learning purposes.

Features of a good HTML editor are:

- **Syntax highlighting:** This feature separates HTML tags into different colors based on their categories, making it easier to read and recognize the code structure.
- **Auto-completion:** This feature automatically suggests names and attributes of an HTML element based on previously added values, saving time when typing a long piece of code.
- **Error detection:** This feature scans the code for syntax errors, and whenever you type the wrong code, it provides options to correct it immediately.
- **Search and replace:** This is a feature to find and replace texts in a document or files in a folder, which helps in saving time for searching and replacing.
- **Code folding:** This feature hides some parts of the code and helps the developer focus on certain parts of the document.
- **FTP integration:** This feature connects your editor directly to the webserver via an FTP client.

Assignment -

Advantages and disadvantages of HTML editors

Advantages -

HTML editors have significant importance because these types of software help users systematically check code snippets. These editors offer different features such that they can insert some of the commonly used HTML tags and structures within the document, and they also provide code auto-complete features. Later, users can translate the code generated through an HTML editor into different languages such as XML, JavaScript, etc. For example, the NVU editor has this translation functionality. The editors provide a user-friendly and engaging design experience. If users use the benefits of online HTML editors, it is possible to create websites quickly and with high development speed. The HTML editors also provide complete control to the developers, thus allowing them to delve more in-depth into the source code and discover the hidden complications in the HTML codes.

Types of HTML editors -

HTML editors are of two different types. One is a **text-based** HTML editor, and the other is the **WYSIWYG** editor.

- **Text-based HTML editor**
This HTML editor design considers that the user is familiar with HTML and understands how to use the different tags in the HTML. A good text-based editor also offers a correction feature, highlighting syntax errors in different colors. Users can use much online and offline software.

- **WYSIWYG editor (what you see is what you get)**

A WYSIWYG editor is convenient for HTML users who have full or no HTML knowledge. WYSIWYG is short for what you see is what you get. It accurately reflects what an HTML editor can do. Using this editor, users can edit their HTML syntax in the format of the respective websites. The editor can instantly convert all accesses to CSS or HTML. These functions are similar to the prevailing office programs. If users make any changes to the text or images, these will appear immediately after the user has edited. In addition, users can use the WYSIWYG editor for CMS and blog systems, for example, Joomla and WordPress.

HTML Converters -

An HTML converter is a software program that, in simple terms, converts a basic text file into HTML code. HTML is the universally accepted standard language in which Web pages are written. Documents with formats such as .doc, .docx and .txt are often created to be posted on the Internet, and the HTML converter aids in this process.

An HTML converter, as the name suggests, is a program developed to accommodate the increasing demand for an easy way to post files of various formats onto websites. An HTML converter is lightweight software developed with the sole purpose of converting documents (mostly text or PDF) into HTML code and preserving their original formatting, so that they can then be easily uploaded to the Internet. HTML converters have significantly helped amateurs and non-professionals to quickly and successfully accomplish text-to-HTML conversion.

4.2. HTML Editors and tools

4.2.1. Use of different HTML editors and tools like Dreamweaver, Microsoft Front Page, Notepad++, etc.

<https://data-flair.training/blogs/html-editors/>

Assignment –

Write the uses of HTML editors.

HTML editor is a software used for writing code in HTML, which is used for structuring and creating websites. Even though codes can be written from scratch using a normal text editor, HTML editors provide a great deal of ease to the developers by ensuring hassle-free coding.

When should you use HTML Editor?

1. Developers prefer to use HTML editors when they want to have a full control over their code and easily create their websites.

2. HTML editors are of great importance to the users who don't have much knowledge of HTML, as of now, and also those who need to generate source codes quickly.
3. HTML editors are highly beneficial for the sake of convenience as they successfully conceal and correct the developers' part of minor errors by syntax correction, auto-completion, simple editing, etc.

Some List of HTML editors:

Many specialized software applications are available online for editing HTML code and creating web pages. Here is a list of some popular HTML editors:

- Notepad
- TextEdit
- Sublime Text
- Atom
- Notepad ++
- Visual Studio Code
- Adobe Dreamweaver
- Google Web Designer
- Bluefish
- Brackets
- NetBeans

Dreamweaver

Adobe Dreamweaver is a proprietary web development tool from Adobe Inc. It was created by Macromedia in 1997 and developed by them until Macromedia was acquired by Adobe Systems in 2005. Adobe Dreamweaver is available for the macOS and Windows operating systems.

<https://www.du.edu/it/services/software/software-catalog/395771a01b1ea850689210e58d4bcbdb6#:~:text=What%20is%20Dreamweaver%3F,from%20beginners%20to%20advanced%20professionals.>

Microsoft Front Page HTML Tool –

<https://web.ics.purdue.edu/~pbawa/421/Introduction%20to%20Front%20Page.htm>

Microsoft FrontPage is a discontinued WYSIWYG HTML editor and website administration tool from Microsoft for the Microsoft Windows line of operating systems. It was branded as part of the Microsoft Office suite from 1997 to 2003.

Notepad++

Notepad++ is a free and open-source text and source code editor for use with Microsoft Windows. It supports tabbed editing, which allows working with multiple open files in a single window. Notepad++ is distributed as free software.

<https://notepad-plus-plus.org/>

4.3. Graphical and Animation Tools

4.3.1. Use of different graphical and animation tools like Adobe Photoshop, MS Paint, Flash, etc.

4.3.2. Adding Sounds and Animation to the web page (using embed tag)

<https://filmlifestyle.com/best-animation-tools/>

<https://blog.appsumo.com/best-animation-software/>

<https://codecondo.com/tools-for-creating-animations-with-html5/>

Graphical and Animation tools -

Graphics software is a type of computer program that is used to create and edit images. There is a wide range of graphics software available on the market, ranging from simple programs that allow users to create and edit basic images, to complex tools that can be used to create detailed 3D models and animations. Some of the most popular graphics software programs include Adobe Photoshop, Corel Painter, and Autodesk Maya.

Animation software: This is the software that you use to create animations (moving images) either by animating existing graphics or by creating new ones from scratch. Common animation software includes Flash, Adobe After Effects, Apple Motion, and Autodesk Maya. Animation software is often used for movies, commercials, and video games.

Extra –

Difference between graphics and animation

<https://anydifferencebetween.com/difference-between-graphics-and-animation/>

4.3.1. Use of different graphical and animation tools like Adobe Photoshop, MS Paint, Flash, etc.

<https://www.geeksforgeeks.org/what-is-graphics-software/>

Assignment –

Write Uses of graphic and animation tools in html

4.3.2. Adding Sounds and Animation to the web page (using embed tag)

The **<embed> tag** in HTML is used to embed external resources, such as videos, games, or audio files, into an HTML document. It is an empty tag, meaning it does not have a closing tag and can't contain any content.

Example –

```
<!-- HTML code for a webpage with an embedded video -->
<html>
  <head>
    <title>My Video</title>
  </head>
  <body>
    <h1>My Video</h1>
    <embed src="video.mp4" type="video/mp4" />
  </body>
</html>
```

In the above example, The src attribute specifies the URL of the resource to be embedded, and the type attribute specifies the MIME type of the resource. In this example, we specify that the resource is a video in MP4 format.

The <embed> tag can also embed other types of resources, such as audio files, Flash animations, PDFs, etc.

Example:

```
<!-- Embed an audio file -->
<embed src="audiofile.mp3" type="audio/mpeg">

<!-- Embed a Flash game -->
<embed src="game.swf" type="application/x-shockwave-flash">

<!-- Embed a PDF -->
<embed src="document.pdf" type="application/pdf" width="800" height="600" />
```

In addition to the src and type attributes, the <embed> tag also supports a number of other attributes that can be used to customize the appearance and behavior of the embedded resource. For example, the width and height attributes can specify the dimensions of the embedded resource, or the autoplay attribute can be used to play the resource on page load automatically.

Here is an example of how to use some of these additional attributes:

Example:

```
<!-- Embed a video with custom dimensions and autoplay -->
```

```
<embed src="myvideo.mp4" type="video/mp4" width="640" height="360" autoplay>
```

Note -

Not all browsers support the `<embed>` tag, so alternative methods can be used to embed external resources in HTML documents. For example, the `<iframe>` tag can be used to embed external web pages, or the `<video>` and `<audio>` tags can be used to embed multimedia files.